

Cortisol

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Establish the diagnosis of adrenocortical insufficiency, Addison's disease, adrenocortical hypersecretion, Cushing's syndrome. Malfunction of the organs in the hypothalamic-pituitary-adrenal cortex axis will result in alteration of the cortisol levels.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Pregnancy, contraceptives, and estrogen therapy give rise to elevated cortisol concentrations.¹ In samples from patients who have been treated with prednisolone, methylprednisolone, or prednisone, falsely elevated concentrations of cortisol may be determined.¹ During metyrapone tests, 11-deoxycortisol levels are elevated.¹ Falsely-elevated cortisol values may be determined due to cross reactions. Patients suffering from 21-hydroxylase-deficiency exhibit elevated 21-deoxycortisol levels and this can also give rise to elevated cortisol levels.¹

The time of sample collection must be taken into account when interpreting results due to the cortisol secretion circadian rhythm. Severe stress can also give rise to elevated cortisol levels.¹

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.¹ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.¹ The test contains additives, which minimize these effects.

The determination of cortisol in 24-hour urine is the method of choice for the detection of Cushing syndrome since cortisol excretion in urine is not subject to the diurnal rhythm of cortisol secretion.²

Custom Additional Information

Cortisol (hydrocortisone) is the most prominent glucocorticosteroid, and it is essential for the maintenance of several body functions. Like other glucocorticosteroids, cortisol is synthesized from the common precursor cholesterol in the zona fasciculata of the cortex of the adrenal gland. For the transport of cortisol in blood, about 90% of cortisol is bound to corticosteroid-binding globulin (CBG) and to albumin. Only a small amount of cortisol circulates unbound in blood and is free to interact with its receptors.³

The most important physiological effects of cortisol are the increase of blood glucose levels (enhancement of gluconeogenesis, catabolic action) and its anti-inflammatory and immunosuppressive action.³

Synthesis and secretion of cortisol by the adrenal gland are controlled by a negative feedback mechanism within the hypothalamus-pituitary-adrenal cortex-axis. If the cortisol level is low, corticotropin-releasing hormone (CRH) is secreted by the hypothalamus, which causes the pituitary to release adrenocorticotrophic hormone (ACTH). This stimulates the synthesis and secretion of cortisol by the adrenal gland. Cortisol itself acts in a negative feedback mechanism on the pituitary gland and the hypothalamus. In addition, stress is followed by increased cortisol secretion.³

Serum cortisol concentrations normally show a diurnal variation.³

Maximum concentrations are usually reached early in the morning and then concentrations decline throughout the day to an evening level that is about half of the morning concentration; therefore, for interpretation of results, it is important to know the collection time of the serum sample.

The cortisol status of a patient is used to diagnose the function or malfunction of the adrenal gland, the pituitary, and the hypothalamus.^{4,5} Thereby, cortisol serum concentrations are used for monitoring several diseases with an overproduction (eg, Cushing syndrome)^{6,7} or underproduction (eg, Addison disease) of cortisol and for monitoring several therapeutic approaches (eg, dexamethasone suppression therapy in Cushing syndrome and hormone replacement therapy in Addison disease).

Specimen Requirements

Specimen

Serum

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube (preferred) or red-top tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube. Blood should be drawn at 8 AM and 4 PM to evaluate baseline diurnal variation (see **Cortisol, AM & PM [104000]** (/tests/104000/cortisol-am-amp-pm)).

Morning specimen is often ordered with ACTH level.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

AM: 6.2–19.4 µg/dL; PM: 2.3–11.9 µg/dL

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

References

Bravo EL. Physiology of the adrenal cortex. *Urol Clin North Am*. 1989 Aug; 16(3):433-437 (review). PubMed 2665269

(<http://www.ncbi.nlm.nih.gov/pubmed/2665269>)

Kirkman S, Nelson DH. Alcohol-induced pseudo-Cushing's disease: A study of prevalence with review of the literature. *Metabolism*. 1988 apr; 37(4):390-394 (review). PubMed 2833680

(<http://www.ncbi.nlm.nih.gov/pubmed/2833680>)

Footnotes

1. Cortisol (Serum) on Elecsys 1010/2010 and Modular Analytics E170. Package insert 2005-02, V 10. Indianapolis, Ind: Roche Diagnostics; 2005.

2. Miyachi Y. Pathophysiology and diagnosis of Cushing's syndrome. *Biomed Pharmacother.* 2000 Jun; 54(Suppl 1):S113-117. PubMed 10915006 (<http://www.ncbi.nlm.nih.gov/pubmed/10915006>)
3. Aron DC, Tyrell JB. Glucocorticoids and adrenal androgens. In: Greenspan FS, Baxter JD, eds. *Basic and Clinical Endocrinology*. 4th ed. Norwalk, Conn: Appleton and Lange;1994:307-346.
4. Hasinski S. Assessment of adrenal glucocorticoid function. Which tests are appropriate for screening? *Postgrad Med.* 1998 Jul; 104(1):69-72. PubMed 9676562 (<http://www.ncbi.nlm.nih.gov/pubmed/9676562>)
5. Rosalki SB. Biochemical testing of adrenocortical function. *Int J Clin Pract.* 1998 Apr-May; 52(3):189-191. PubMed 9684436 (<http://www.ncbi.nlm.nih.gov/pubmed/9684436>)
6. Newell-Price J, Trainer P, Besser M, Grossman A. The diagnosis and differential diagnosis of Cushing's syndrome and pseudo-Cushing's states. *Endocr Rev.* 1998 Oct; 19(5):647-672. PubMed 9793762 (<http://www.ncbi.nlm.nih.gov/pubmed/9793762>)
7. Ross RJ, Trainer PJ. Endocrine investigation: Cushing's syndrome. *Clin Endocrinol (Oxf).* 1998 Aug; 49(2):153-5. PubMed 9828897 (<http://www.ncbi.nlm.nih.gov/pubmed/9828897>)